

Equivariant Spaces

Labix

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Abstract

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1 Topological Groups and G-Spaces

The above four chapters has established deep connections between three properties of a space. Namely, the fundamental group, the fibers of the covering space and the group of homeomorphisms of the covering space. Such a deep connection between algebra and topology is not unique to covering space theory, nor to fundamental groups. In this section we will take a step back and look at the big picture.

1.1 Topological Groups and the Coset Space

Definition 1.1.1 (Topological Groups) Let G be a group. We say that G is a topological group if G is also a topological space and that the following are true.

- The multiplication map $\cdot : G \times G \rightarrow G$ defined by $(g, h) \mapsto gh$ is continuous.
- The inverse map $(-)^{-1} : G \rightarrow G$ defined by $g \mapsto g^{-1}$ is continuous.

Notice that every group can be given the discrete topology, and so every group is trivially a topological group. But of course there is no guarantee that anything interesting theorems will occur in this case. We call these topological groups discrete.

Definition 1.1.2 (Discrete Group) Let G be a topological group. We say that G is a discrete group if it has the discrete topology.

Proposition 1.1.3 Let G be a topological group. Let H be a subgroup of G . Then H and $\overline{H}$ are both topological groups. Moreover, if H is normal, then $\overline{H}$ is normal.

Proposition 1.1.4 Let G be a topological group. Let H be a subgroup of G . Then the normalizer $N_G(H)$ and the centralizer $C_G(H)$ are closed subgroups of G .

Definition 1.1.5 (The Coset Space) Let G be a topological group and H a closed subgroup of G . Define the coset space of H in G to be the quotient space

$$G/H = \{gH \mid g \in G\}$$

together with the (topological) quotient map $p : G \rightarrow G/H$ such that $U \subseteq G/H$ is open if and only if $p^{-1}(U)$ is open.

Theorem 1.1.6 Let G be a topological group. Let H be a subgroup of G . Then the (topological) quotient map

$$p : G \rightarrow G/H$$

is an open map. Moreover, the following are true regarding the quotient.

- G/H is Hausdorff if and only if H is closed in G
- G/H is discrete if and only if H is open in G .
- If H is normal and closed in G , then G/H is a topological group.

1.2 Morphisms Between Topological Groups

Definition 1.2.1 (Continuous Homomorphisms) Let G and H be topological groups. A function $f : G \rightarrow H$ is said to be a continuous homomorphism if it is continuous and a group homomorphism.

Proposition 1.2.2 Let G, H be topological groups. Let $\varphi : G \rightarrow H$ be a surjective continuous homomorphism. Then $\ker(\varphi)$ is a closed subgroup of G and φ is a continuous bijection.

When the topological group G is compact, the first isomorphism theorem in fact gives a homeomorphism.

Proposition 1.2.3 Let G, H be topological groups. Let $\varphi : G \rightarrow H$ be a surjective continuous homomorphism. If G is compact, then

$$\bar{\varphi} : \frac{G}{\ker(\varphi)} \rightarrow H$$

is a homeomorphism.

Proposition 1.2.4 Let G be a compact topological group. Let $g \in G$. Then

$$A = \overline{\{g^n \mid n \in \mathbb{N}\}}$$

is a subgroup of G .

1.3 Some Real Topological Groups

Proposition 1.3.1 (The Group of Matrices as a Topological Group) The group of real $n \times n$ matrices

$$M_n(\mathbb{R}) = \{(a_{ij})_{n \times n} \mid a_{ij} \in \mathbb{R}\}$$

with group structure given by matrix addition can be given the structure of a topological group whose topology is given by the topology of $\mathbb{R}^{n^2}$ and a choice of bijection of sets $\mathbb{R}^{n^2} \cong M_n(\mathbb{R})$.

Proposition 1.3.2 Let $n \in \mathbb{N} \setminus \{0\}$. Then $GL(n, \mathbb{R})$ is an open subset of $M_n(\mathbb{R})$.

Beware that $GL(n, \mathbb{R}), O(n, \mathbb{R}), SL(n, \mathbb{R}), SO(n, \mathbb{R})$ are not topological groups as subsets of $M_n(\mathbb{R})$ because they are not subgroups of $M_n(\mathbb{R})$ in the first place.

Proposition 1.3.3 (The General Linear Group as a Topological Group) The general linear group

$$GL(n, \mathbb{R}) = \{M \in M_n(\mathbb{R}) \mid \det(M) \neq 0\}$$

with group structure given by matrix multiplication can be given the structure of a topological group whose topology is given by the subspace topology as a subset of $M_n(\mathbb{R})$.

Proposition 1.3.4 Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are all topological groups with group structure and topology inherited from $GL(n, \mathbb{R})$.

- The orthogonal group $O(n, \mathbb{R})$.
- The special linear group $SL(n, \mathbb{R})$.
- The special orthogonal group $SO(n, \mathbb{R})$.

Moreover, they are closed subsets of $GL(n, \mathbb{R})$.

1.4 Some Complex Topological Groups

$GL(n, \mathbb{C}), U(n), SU(n)$

2 Equivariant Spaces

2.1 G-Spaces and G-Equivariant Maps

In algebraic topology, we have the results of considering groups acting on spaces. We can in fact consider topological groups acting on spaces.

Definition 2.1.1 (Continuous Group Actions) Let G be a topological group and X a space. We say that G is a continuous group action if G is a group acting on X such that the group action map

$$\cdot : G \times X \rightarrow X$$

is continuous. In this case we say that X is a G -space.

Frequently a continuous group action is also called a (topological) transformation group, for example in Milnor's Topology of Fiber Bundles or Introduction to Compact Topological Groups.

Proposition 2.1.2 Let G be a continuous group action of X . Then for each $g \in G$, the left action map $x \mapsto g \cdot x$ is a homeomorphism of X .

Proof Every element of g has an inverse g^{-1} which are both continuous and are bijections on X . ■

Proposition 2.1.3 Let G be a topological group and $(X, \mathcal{T})$ a topological space. Then G is a continuous group action on X if and only if G acts on $\mathcal{T}$.

Proof Suppose that G is a continuous group action on X . Then for each $g \in G$, $g \cdot U = \{g \cdot x \mid x \in U\}$ for $U \in \mathcal{T}$ is open since A_g as above is a homeomorphism. Now suppose that G acts on $\mathcal{T}$. Then for each open set U of X , $g^{-1} \cdot U$ is open. Thus G is a continuous group action. ■

In particular, some authors would assume one knows this fact, so it is always nice to see it spelled out. It is also standard to denote this action just by the element g instead of A_g .

Definition 2.1.4 (Group of Homeomorphisms) Let X be a space. Define the group of homeomorphisms of X to be

$$\text{Homeo}(X) = \{f : X \rightarrow X \mid f \text{ is a homeomorphism}\}$$

together with composition of functions. We say that a group A is a subgroup of homeomorphisms of X if A is isomorphic to a subgroup of $\text{Homeo}(X)$.

Lemma 2.1.5 Let G be a topological group. Let X be a G -space. Then there is a group homomorphism $\varphi : G \rightarrow \text{Homeo}(X)$ defined by

$$g \mapsto (x \mapsto g \cdot x)$$

Proof We have already seen that for any $g \in G$, the map $x \mapsto g \cdot x$ is a homeomorphism. Thus the above mapping is well defined. Now we have that

$$\begin{aligned} \varphi(gh)(x) &= gh \cdot x \\ &= g \cdot (h \cdot x) \\ &= (\varphi(g) \circ \varphi(h))(x) \end{aligned}$$

and so φ is a group homomorphism. ■

Notice that if the above group homomorphism is injective, then the structure group G is a subgroup of homeomorphisms of G .

Definition 2.1.6 (G-Equivariant Maps) Let G be a topological group and let X, Y be G -spaces. A G -equivariant map is a continuous map $f : X \rightarrow Y$ such that f is equivariant. In other words, we require that

$$f(g \cdot x) = g \cdot f(x)$$

for all $x \in X$ and all $g \in G$.

Definition 2.1.7 (Isomorphic G-Spaces) Let G be a topological group and let X, Y be G -space. We say that G and H are isomorphic G -spaces if there exists a G -equivariant map such that f is a homeomorphism.

Theorem 2.1.8 Let G be a topological group and let X be a G -space. Then the map

$$p : \frac{G}{\text{Stab}_G(x_0)} \rightarrow Gx_0 \subseteq X$$

induced by the map $g \mapsto g \cdot x_0$ is well defined. Moreover, it is isomorphic to the left G -space Gx_0 .

Proof To show that it is well defined, we want to show that if $g \in \text{Stab}_G(x_0)$, then $g \cdot x_0 = x_0$. But this is true by definition of the stabilizer. By definition of the induced map, it is continuous. Also, the orbit of x_0 is precisely Gx_0 and hence p is a bijection. It remains to show that p is an open map.

To show isomorphism, we also need to show that p is a G -equivariant map. We have that

$$\begin{aligned} p(g \cdot (h\text{Stab}_G(x_0))) &= p(gh\text{Stab}_G(x_0)) \\ &= (gh) \cdot x_0 \\ &= g \cdot (h \cdot x_0) \\ &= g \cdot p(h\text{Stab}_G(x_0)) \end{aligned}$$

so that p is G -equivariant. ■

Definition 2.1.9 (The Category of G-Spaces) Let G be a topological space. Define the category of G -spaces

$${}_G\mathbf{Top}$$

to consist of the following data.

- The objects are the G -spaces
- The morphisms are the G -equivariant spaces
- Composition is given by the composition of functions.

There is an obvious forgetful functor ${}_G\mathbf{Top} \rightarrow \mathbf{Top}$. One of its adjoint should assign the space to a trivial G -action.

Definition 2.1.10 (The Trivial G-Space Functor) Let G be a topological group. Define the trivial G -space functor

$$\text{Triv} : \mathbf{Top} \rightarrow {}_G\mathbf{Top}$$

by the following.

- For each space X , define a group action on X by $g \cdot x = x$ for all $g \in G$ and $x \in X$.
- For each map $f : X \rightarrow Y$, $\text{Triv}(f) = f$ because f is trivially equivariant.

2.2 Induced and Restricted G-Spaces

Definition 2.2.1 (Induced G-Spaces) Let G be a topological group. Let $H \leq G$ be a subgroup. Let X be an H -space. Define the induced G -space of X to be the space

$$\text{Ind}_H^G X = G \times_H X = \frac{G \times X}{\sim}$$

where the relation is generated by $(g \cdot h, x) \sim (g, h \cdot x)$ for $g \in G, h \in H$ and $x \in X$.

Pushout?

Lemma 2.2.2 Let G be a topological group. Let $H \leq G$ be a subgroup. Let X be an H -space. Then the following are true.

- If $X = *$, then $G \times_H X \cong \frac{G}{H}$.
- If $h \cdot x = x$ for all $h \in H$ (the action of H is trivial), then $G \times_H X \cong \frac{G}{H} \times X$.

Definition 2.2.3 (Restricted G-Spaces) Let G be a topological group. Let $H \leq G$ be a subgroup. Let X be a G -space. Define the restriction

$$\text{Res}_H^G X$$

of X to H to be the space X considered as an H -space by the group action of G .

Proposition 2.2.4 Let G be a topological group. Let $H \leq G$ be a subgroup. Then there is an adjunction

$$\text{Ind}_H^G : {}_H\mathbf{Top} \rightleftarrows {}_G\mathbf{Top} : \text{Res}_H^G$$

This means that there is an isomorphism

$$\text{Hom}_{{}_G\mathbf{Top}}(\text{Ind}_H^G X, Y) \cong \text{Hom}_{{}_H\mathbf{Top}}(X, \text{Res}_H^G Y)$$

that is natural in X and Y .

2.3 Fixed Points and Orbit Spaces

Definition 2.3.1 (The Fixed Points Functor) Let G be a topological group. Define the fixed points functor

$$(-)^G : {}_G\mathbf{Top} \rightarrow \mathbf{Top}$$

by the following.

- For each G -space, X , $X^G = \{x \in X \mid g \cdot x = x \text{ for all } g \in G\}$ is the subset of fixed points of G equipped with the subspace topology.
- For each G -equivariant map $f : X \rightarrow Y$, $(f)^G : X^G \rightarrow Y^G$ is the restriction of f to X^G .

Check: it is well defined.

Proposition 2.3.2 Let G be a topological group. There is an adjunction

$$\text{Triv} : \mathbf{Top} \rightleftarrows {}_G\mathbf{Top} : (-)^G$$

This means that there is an isomorphism

$$\text{Hom}_{{}_G\mathbf{Top}}(\text{Triv } X, Y) \cong \text{Hom}_{\mathbf{Top}}(X, Y^G)$$

that is natural in X and Y .

Definition 2.3.3 (The Orbit Space) Let X be a space and G be a group acting on X . Define the orbit space of X and G to be

$$\frac{X}{G} = \{\text{Orb}_G(x) \mid x \in X\}$$

the set of all orbits of G on X , inherited with the quotient topology of the equivalence relation of orbits.

This has the quotient topology because recall from groups and rings that $\text{Orb}_G(x)$ defines an equivalence relation on X .

2.4 The Pointed Analogue

Definition 2.4.1 (Pointed G -Spaces) Let G be a topological group. A pointed G -space is a space X together with a G -equivariant map $*$ $\rightarrow$ X . We denote it by (X, x_0) where x_0 is the image of the map $*$ $\rightarrow$ X .

Lemma 2.4.2 Let G be a topological group. Let X be a G -space. Let (X, x_0) be a pointed space. Then (X, x_0) is a pointed G -space if and only if x_0 is a G -fixed point of X .

Definition 2.4.3 (Induced Pointed G -Spaces) Let G be a topological group. Let $H \leq G$ be a subgroup. Let (X, x_0) be a pointed H -space. Define the induced pointed G -space of (X, x_0) to be the space

$$\text{Ind}_H^G X = G_+ \wedge_H X = \frac{G_+ \wedge X}{\sim}$$

where the relation is generated by $(g \cdot h, x) \sim (g, h \cdot x)$ for all $g \in G, h \in H$ and $x \in X$.

Pushout?

Proposition 2.4.4 Let G be a topological group. Let $H \leq G$ be a subgroup. Then there is an adjunction

$$\text{Ind}_H^G : {}_H\mathbf{Top}_* \rightleftarrows {}_G\mathbf{Top}_* : \text{Res}_H^G$$

This means that there is an isomorphism

$$\text{Hom}_G(\text{Ind}_H^G X, Y) \cong \text{Hom}_H(X, \text{Res}_H^G Y)$$

that is natural in X and Y .

3 Types of Actions on G-Spaces

3.1 Homogenous G-Spaces

Recall that a group action G on X is said to be transitive if for any $x, y \in X$, there exists $g \in G$ such that $g \cdot x = y$.

Definition 3.1.1 (Homogenous G-Space) Let G be a topological group and let X be a G -space. We say that X is a Homogenous G -space if G acts transitively on X .

Much of the theorem we considered in covering space theory was in fact on homogenous G -spaces. For instance, when $\tilde{X}$ is path connected, prp2.4.7 says that the fibers $p^{-1}(x_0)$ of the covering space is a homogenous $\pi_1(X, x_0)$ -space. The following corollary proves thm 2.4.9

Corollary 3.1.2 Let G be a topological group and let X be a homogenous G -space. Then there is an isomorphism of G -spaces

$$\frac{G}{\text{Stab}_G(x_0)} \cong X$$

induced by the map $g \mapsto g \cdot x_0$.

Proof Since G is transitive on X , we have that $Gx_0 = X$. Hence by the above theorem, we obtain the desired isomorphism. ■

Indeed, if $p : \tilde{X} \rightarrow X$ is a covering space, $p^{-1}(x_0)$ is a homogenous $\pi_1(X, x_0)$ -space and it follows that

$$\frac{\pi_1(X, x_0)}{p_*(\pi_1(\tilde{X}, \tilde{x}_0))} \cong p^{-1}(x_0)$$

Notice that $\text{Stab}_{\pi_1(X, x_0)} = \text{im}(p_*)$ is a non-trivial fact that was proven in prp 2.4.7.

Corollary 3.1.3 Let G be a topological group and let X be a homogenous G -space. If G is more over a free action on X , there is an isomorphism of G -spaces

$$G \cong X$$

given by the map $g \mapsto g \cdot x_0$.

Proof If G is free, then the stabilizer is trivial. By the above corollary, we obtain the desired isomorphism. ■

Theorem 3.1.4 Let G be a topological group and let X be a homogenous G -space. Then the following are true.

- For any $\varphi \in \text{Homeo}(X)$, $x \in X$ and $\varphi(x)$ has the same stabilizers
- If $x, y \in X$ has the same stabilizers, then there exists $\varphi \in \text{Homeo}(X)$ such that $\varphi(x) = y$.

Lemma 3.1.5 Let G be a topological group and let X be a homogenous G -space. Let A be a subgroup of $\text{Homeo}(X)$. Then $A = \text{Homeo}(X)$ if and only if for any $x, y \in X$ such that $\text{Stab}_G(x) = \text{Stab}_G(y)$, there exists $\varphi \in A$ such that $\varphi(x) = y$.

Theorem 3.1.6 Let G be a topological group and let X be a homogenous G -space. Then there is a isomorphism of left G -spaces

$$\frac{N(\text{Stab}_G(x_0))}{\text{Stab}_G(x_0)} \cong \text{Homeo}(X)$$

3.2 Properly Discontinuous Group Actions

Definition 3.2.1 (Proper Group Actions) Let G be a topological group acting continuously on a topological space X . The action is said to be proper if the map $G \times X \rightarrow X \times X$ defined by

$$(g, x) \mapsto (x, g \cdot x)$$

is a proper map.

Definition 3.2.2 (Properly Discontinuous Group Actions) Let G be a group acting on a space X . Then we say that G is a properly discontinuous group action if for every compact set $K \subseteq X$, we have

$$(g \cdot K) \cap K \neq \emptyset$$

for finitely many $g \in G$.

Proposition 3.2.3 Every properly discontinuous group action is a wandering action.

Proposition 3.2.4 If G is a proper group action on a space X , then the action is properly discontinuous.

The converse is not true in general, unless we assume that X is locally compact.

Recall the notion of a covering space action. G is a covering space action on X if $g \cdot U \cap U \neq \emptyset$ implies $g = 1$. This is also related to properly discontinuous group actions. In fact, properly discontinuous group actions are in general stronger than covering space actions.

Proposition 3.2.5 Let G be a covering space action on X . If X is locally compact and Hausdorff, then G is a properly discontinuous group action on X .

3.3 Covering Space Actions

Definition 3.3.1 (Wandering Actions) Let X be a space. Let G be a group acting on X . We say that G is a wandering action on X if for all $x \in X$, there exists a neighbourhood U of x such that

$$(g \cdot U) \cap U \neq \emptyset$$

for finitely many $g \in G$.

Algebraic topologists are primarily interested in the following type of group actions.

Definition 3.3.2 (Covering Space Action) Let X be a space and G be a group acting on X . We say that G is a covering space action if for each $x \in X$, there is a neighbourhood U of x such that

$$(g_1 \cdot U) \cap (g_2 \cdot U) = \emptyset$$

for all $g_1, g_2 \in G$.

Lemma 3.3.3 Every covering space action is wandering and free.

Proof Let G be a covering space action. Then $(g \cdot U) \cap U = \emptyset$ for all $g \in G$ implies that $g \cdot x$ cannot be equal to x . Thus G is a free action on X . It is clear that G is a wandering action on X since we require all actions of $g \in G$ to be disjoint while for wandering actions, we only require a

finite amount of actions of $g \in G$ to be disjoint. ■

The following proposition will show where the name of covering space actions comes from. In particular, we will see that if X is path connected, then there is one unique covering space action on X , namely via the deck group $\text{Deck}(p)$.

Proposition 3.3.4 Let X be a space and $p : \tilde{X} \rightarrow X$ be a covering of X . Then the action of $\text{Deck}(p)$ on $\tilde{X}$ is a covering space action.

Proof Suppose that $\tilde{x} \in (\tau_1 \cdot U) \cap (\tau_2 \cdot U)$. Then this means that $\tau_1(\tilde{x}_1) = \tau_2(\tilde{x}_2)$ for some $\tilde{x}_1, \tilde{x}_2 \in U$. But we have that $p \circ \tau_1 = p \circ \tau_2$ which implies that $\tilde{x}_1$ and $\tilde{x}_2$ lie in the same fiber $p^{-1}(x)$ for some $x \in X$. By definition of covering spaces, the fiber $p^{-1}(x_0)$ intersects U at exactly one point so that $\tilde{x}_1 = \tilde{x}_2$. But this implies that $\tau_2^{-1}\tau_1$ fixes one point in $\tilde{X}$ so that $\tau_2^{-1}\tau_1 = 1$ and $\tau_1 = \tau_2$. ■

Lemma 3.3.5 Let X be a space and let $p : \tilde{X} \rightarrow X$ be a regular covering space of X . Then the orbit space of the deck group

$$\frac{\tilde{X}}{\text{Deck}(p)} \cong X$$

is isomorphic to the base space.

Proposition 3.3.6 Let X be a space and let G be a covering space action on X . Then the quotient map $p : X \rightarrow X/G$ defined by $p(x) = \text{Orb}_G(x)$ is a regular covering space of X/G .

Theorem 3.3.7 Let X be a path connected space. Let G be a covering space action on X . Then $G \cong \text{Deck}(p)$ where $p : X \rightarrow X/G$ is the regular covering space of X/G .

Corollary 3.3.8 Let X be a path connected and locally path connected space. Then there is a group isomorphism

$$\text{Deck}(p) \cong \frac{\pi_1 \left(\frac{X}{\text{Deck}(p)}, x_0 \right)}{p_* (\pi_1 (X, p(x_0)))}$$

for any $x_0 \in X$.

4 Equivariant Homotopy Theory

4.1 G-Homotopy

Definition 4.1.1 (G-Homotopy) Let G be a topological group and let X, Y be G -spaces. Let $f, g : X \rightarrow Y$ be G -equivariant maps. A G -homotopy from f to g is a homotopy $H : X \times I \rightarrow Y$ from f to g such that for each $t \in I$, the map

$$H(-, t) : X \rightarrow Y$$

is G -equivariant map.